

The MyGTAP Single Household Program

Instructions and documentation

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1 What is the MyGTAP Single Household Program?

With increasing interest in the basic features of the MyGTAP model, the decision was made to divide the MyGTAP framework into two parts. This will provide users with access to the more basic features of the MyGTAP model without having to disaggregate the households or factors of a specific country. The MyGTAP framework will therefore consist of two parts:

- The MyGTAP single household program that extends the standard GTAP database to include foreign income flows, such as remittances, foreign capital income, foreign aid, and government transfers for all GTAP countries; and removes the GTAP regional household.
- The MyGTAP multiple household program that allows the user to add additional households and factors for a specific country.

This document focuses on the MyGTAP single household program. The MyGTAP single household program can be loaded into a RunGTAP application with a GTAP aggregation to convert it into a ready to use MyGTAP model application, thereby allowing users to use the MyGTAP model as they would the GTAP model.

The benefits of using the MyGTAP model in this mode are: 1) the MyGTAP model captures additional foreign income flows that affect the current account and income, thereby improving the distinction between (and hence the analysis of) gross domestic product (GDP) and gross national income (GNI); 2) the inclusion of these additional data and corresponding variables provides the user with a number of additional policy instruments that can be examine (for example, the user can examine the impact of changes in aid payments or remittance flows); and 3) the MyGTAP model removes the regional household, directly linking government income and expenditures on the one hand, and private household income and expenditures on the other; thereby providing the user with a number of additional closure choices.

This brief instruction guide accompanies the single household MyGTAP program. The document describes the data sources used to extend the GTAP 9a database (Narayanan, Aguiar and McDougall, 2015) in section 2; and provides details and instructions on how to run the program in section 3. Users interested in complete documentation of the MyGTAP model and database, and in particular the ability to incorporate additional households and factors, are referred to Minor and Walmsley (2013a) and Walmsley and Minor (2013b).

2 Sources of the additional data used in MyGTAP

The MyGTAP database program contains three additional pieces of data: foreign capital income; remittances; and foreign aid; as well as a placeholder for government-household transfers. The sources of these data are discussed in turn below:

The IMF's Balance of Payments Statistics (IMF BOPS, 2015)¹ database (<http://www.imf.org/en/Data#data>) is used to provide data on foreign income transactions (remittances and capital income).

- Foreign capital income (finc.har) includes Income, Investment Income, Debit/credit (BPM5)
- Remittances (frem.har) include Income, Compensation of Employees, Debit/credit (BPM5) and Income, Current Transfers, Other Sectors, Workers' Remittances, Debit (BPM5)

Foreign aid is obtained from a number of sources, including the IMF's Balance of Payments Statistics (IMF BOPS, 2015) database, the IMF's Government Financial Statistics (IMF GFS, 2015) database (<http://www.imf.org/en/Data#data>)² and the OECD's Development Assistance Committee (DAC) database (OECD, 2015). Since the data are not consistent between the sources we use the maximum inflows and outflows from each of the three alternative sources.

Finally, transfers are set to zero in the database construction program, except in certain circumstances explained below. They can be edited by the user, if the user has access to better data. There are some circumstances where a value from the IMF's Government Financial Statistics (IMF GFS, 2015) has been used – the sum of other expenses, interest payments and social security expenses less social security revenues and other revenues. Due to issues with the IMF data, these data were not always used. The following conditions had to be met before they were used: 1) the IMF GFS (2015) data must include both revenue and expense data for to obtain a net transfer figure; 2) there must be total revenue and expenses and hence have a government balance estimate; and 3) the inclusion of these transfers must move the GTAP government balance closer to the government balance reported in the IMF GFS (2015). In many countries these conditions were not met and hence the IMF GFS data were not used.

3 Instructions: how do I create a MyGTAP application in RunGTAP?

In the following sections step-by step instructions are provided on how to aggregate the GTAP Database and place it in RunGTAP and then convert it into a MyGTAP application ready to run a simulation. An overview of those steps is given below or find the instructions most relevant to you:

- What software do I need? (section 3.1)
- Run the GTAPAgg program to create the GTAP database (section 3.2)

¹ See IMF BOPS (1993) for documentation.

² See IMF GFS (2014) for documentation of this database.

- Loading MyGTAP model and GTAP data into RunGTAP (section 3.3)

3.1 What software do I need?

The MyGTAP single household program requires the GTAP 9a Data Base aggregated using GTAPAg2. The GTAP 9a Data Base includes data for 140 countries and 57 sectors and eight factors of production. The GTAP Data Base must be purchased from the GTAP Center and a licensed copy is required. We will assume that the GTAPAg2 program is already installed on your computer.

The MyGTAP single household program is available from ImpactECON as a ZIP file. The first section of this step-by-step process includes direction for setting up the GTAP Data Base files.

The RunGTAP software must be installed on your machine. RunGTAP can be obtained from the GTAP website at:

<https://www.gtap.agecon.purdue.edu/products/rungtap/default.asp>

Depending on the size of your aggregation and the extent to which you wish to change the model, you may also need a Gempack license. More details can be found at:

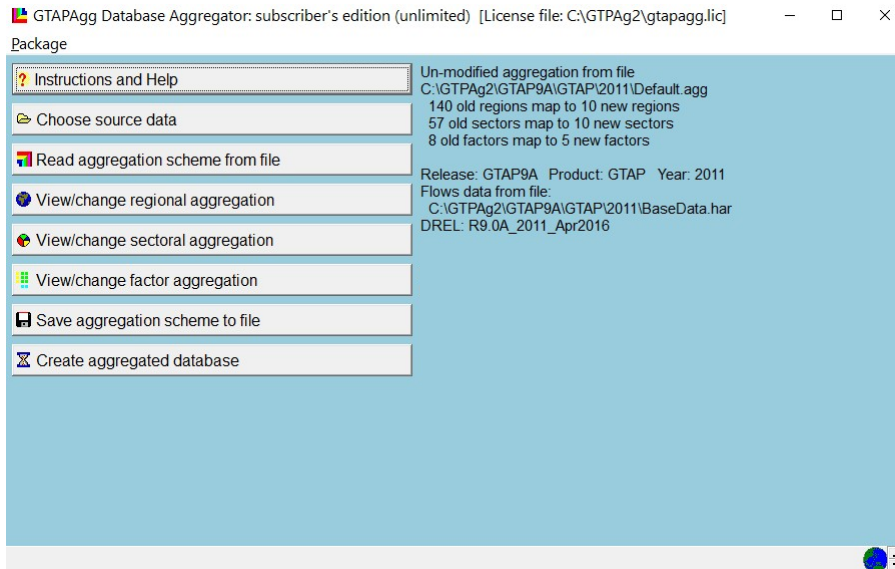
<http://www.copsmodels.com/gemfeat.htm>

You will also need to be able to copy and paste files in something like file explorer; and extract files from a zip file again. This should not require any additional software, although you may have your own preferences or programs for undertaking these tasks.

3.2 Run the GTAPAg2 program to create the GTAP database

Our first task is to create an aggregation of the GTAP Data Base. To do this, we use GTAPAg2:

- **Open GTAPAg2** by doubling clicking on the GTAPAg2 icon on your desktop.



The buttons in GTAPAg2 read like a to-do list, so you naturally move from one button to the next.

First, make sure that you have loaded in the correct package by selecting package from the main menu and navigating to the GTAP9a database. Once the GTAP9A package is loaded you can then choose the specific reference year by selecting “Choose source data”. In this case, we will be employing the V9a 2011 database, so you should select v9a and 2011. The information area of the screen, below the previous text, should reference the 2011 database. For instance, our GTAPAg2 references DREL: R9.0A_2011_Apr2015.

GTAPAg2 is a flexible program, allowing the user to select sectors and countries\regions for aggregation. However, given the large number of countries, it is often helpful to start with an aggregation file. In the GTAPAgg directory, there is usually a “default.agg” file, which provides a 10 region by 10 sector aggregation, and is automatically loaded up into the program.

The regional aggregation can then be altered by:

Click on “View/change regional aggregation”.

Here you can review and change the aggregation. The screen has three 3 main areas: a list of all 140 regions and their mapping (upper right); buttons to change the mapping (to the bottom left of the regional mapping); at the bottom, there is a list of the aggregated regions, and the corresponding GTAP regions along with a new description of the aggregated sector.

- Adding a region requires you to right click in the bottom window and select “insert before” or select “insert after”. A new row will be created, but will not have any useful data.
- Add a new region code (short meaningful names of no more than 8 characters are best) and new region description.

- Return to the 140 regions in the upper right-navigate to the country or region you wish to assign this new region.
- In the “New Region” column, click in the region, you will be presented with a list of the regions contained in the lower pane, including the new region you just created. Select the new region (or skip this if you are just following along and do not wish to change the aggregation).
- Click “OK” to return to the main GTAPAg2 screen.

The sectors and factor aggregations can be changed in the same way by clicking on the appropriate button (“View/change sectoral aggregation” or “View/change factor aggregation” respectively).

Once you have your preferred aggregation:

Click on **“Save aggregation scheme to file”**.

Choose a new name and location for the aggregation file.

Then Click on **“Create aggregated database** in order to create the new aggregated database.

You may be prompted to save the aggregation and/or assign a location for the aggregated data, which will be saved in a zip file by default.

Choose a location and file name you will remember, perhaps c:\rungtap\archive.

Click **“save”**.

You can now view the output files if you like by clicking on “view output files”.

That’s it, you have done it!

3.3 Loading MyGTAP model and data files into RunGTAP

Once you have created your GTAP aggregation you need to put it under a new directory in the RunGTAP directory so you can start running your simulations. This section assumes you have already installed the RunGTAP software on your computer and that you have the appropriate GEMPACK license for your aggregation.

The first step is to **create a directory** for the model and database under RunGTAP. To do this:

- Navigate to the RunGTAP directory containing the RunGTAP application.
- This should be in a directory like “c:\RunGTAP5”.
- Create a new directory for the new model and data files, call this directory “Myapp”.

e.g., c:\RunGTAP5\MyApp

You may use whatever name you like, but make sure it is no more than 8 characters long. This is also the name that will be given to your version in RunGTAP.

The second step is to **load the MyGTAP program files** into your RunGTAP directory:

- You will have obtained these in the zip file you downloaded from the ImpactECON website: <https://impactecon.com/>. They are the files contained in the folder rungtapsh32 in this zip file.
- **Unzip** the files from this folder into the directory you created in the first step

c:\RunGTAP5\Myapp

Step three is to add **your new GTAP data files to the application or directory**:

- Navigate to the zip file containing the aggregated data created in section 3.2. The suggested directory was c:\runGTAP\archive.
- **Extract** the zip file into the new directory created in runGTAP. c:\rungtap5\Myapp.

Note: do NOT load up the data using the RunGTAP version | new features. If you do this the RunGTAP module settings will be out and you will need to adjust them manually.

Step four is to **run the MyGTAP batch file**:

- In Windows explorer, navigate to the MyGTAP directory you just created ("myapp" if you are following along here).
- Double click the batch file named

"MyGTAPshaux.bat"

- You will be asked two questions:
 - o What version of the GTAP database are you using? If you are using v9.0a, type 9a. As more versions of the GTAP database become available we will update the program to accept new data.
 - o What is the base year of your data? Currently you can use 2007 or 2011 data. Type 2011 or 2007 depending on the version of your data.

You can now open up RunGTAP and navigate to your new version. To do this:

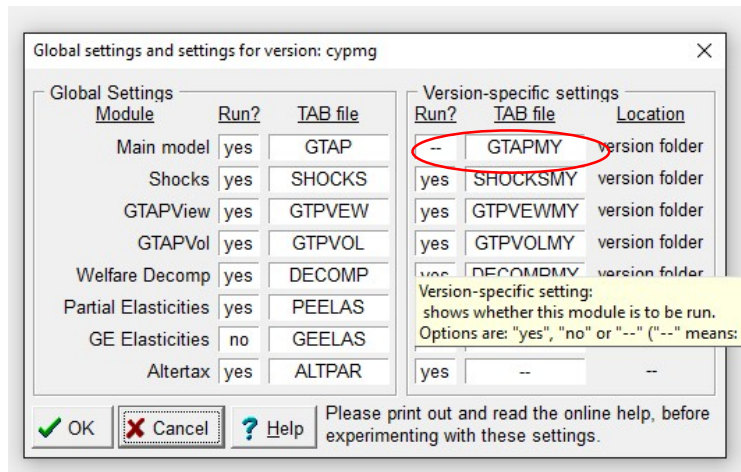
- Open RunGTAP, by double clicking on the RunGTAP icon on your desktop
- In the menu, select "Version | Change".
- Your new directory containing the model files should be included in the list of versions.
- Choose the version you created in the earlier if you are following along this should be "Myapp".
- Click "OK".

Finally, we will **check** that you have the correct data and model and that it runs:

- First let's check the data.
 - o The core database files (sets.har, basedata.har and default.prm) now contain two special headers – MREL and DREL. MREL is the MyGTAP version and

DREL is the GTAP region. Both headers should refer to the same version, v9a, and the same year, 2011 or 2007 as appropriate. If one of these headers refers to 2011 and the other 2007, then you have used the wrong data and you need to go back to section 3.2 or the beginning of section 3.3 depending on which data are incorrect – GTAP (DREL) or MyGTAP (MREL) respectively.³

- Close the header array file and return to the RunGTAP interface.
- Second, let's check the correct model is loaded.
 - Click on “Version | modules” in the main menu.
 - The following box should appear:



- If you can see GTAPMy listed in the version-specific settings on the right-hand side (circled in red) then you are using the correct model.
- If not, you can click on this area and navigate to the RunGTAP/myapp directory and find the GTAPMy.tab file. If you get any errors, then you need to go back to step 2.
- Finally let's check the model solves
 - **Click on the “Solve” tab.**
 - The numeraire shock should be loaded. To check look at the shocks page and confirm that you are shocking pfactwld – the global numeraire.
 - Then return to the solve tab and

Click on “Solve”

- If the model solves, you have done it! You can take a look to ensure that the model passed the homogeneity test. If it does not, then please contact us.

³ If you wish to convert files for another year, just repeat steps 3.2 and 3.3 making sure that you use the 2007 GTAP data and the 2007 MyGTAP inputs.

Congratulations, the model and database are installed correctly and you are ready to build your shocks for a real simulation.

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